

Students' approach to the use of generative AI

Ambassador: Vidminas Vizgirda, School of Informatics – vidminas.github.io
Host: Jordana Viotto Da Cruz, Business School



Task

192 UG and MSc students taking the Digital Business course participated in a competition.

Task 1: "You have just joined a team of three entrepreneurs who will start a business together. All the team has decided so far is that you will start a digital platform that facilitates transactions or interactions among users. Your team wants to enter a competition where you can get £100,000 in seed funding. Your role is to write a business proposal with the potential to win the competition. In this competition, business proposals will be evaluated according to their creativity."



Task 2: "Now you have the opportunity to consult ChatGPT before submitting your final proposal. You will consult ChatGPT using at least 2 prompts*, in the ways you consider the best to successfully finish this task. (*Prompts are instructions given to artificial intelligence.) Once you are satisfied, you will choose one of the three options below. Please choose the option that gives you the highest chance of winning the competition."

Which proposals were more original? 🤔🤔🤔

Data

- 192 paired texts submitted by students (for task 1 and task 2)
- Texts ranged in length from 144 words to 16600 words – so the dataset included both very short and very long texts
- Median text length for task 1 was 966.5 words and for task 2 – 1661.5 words.

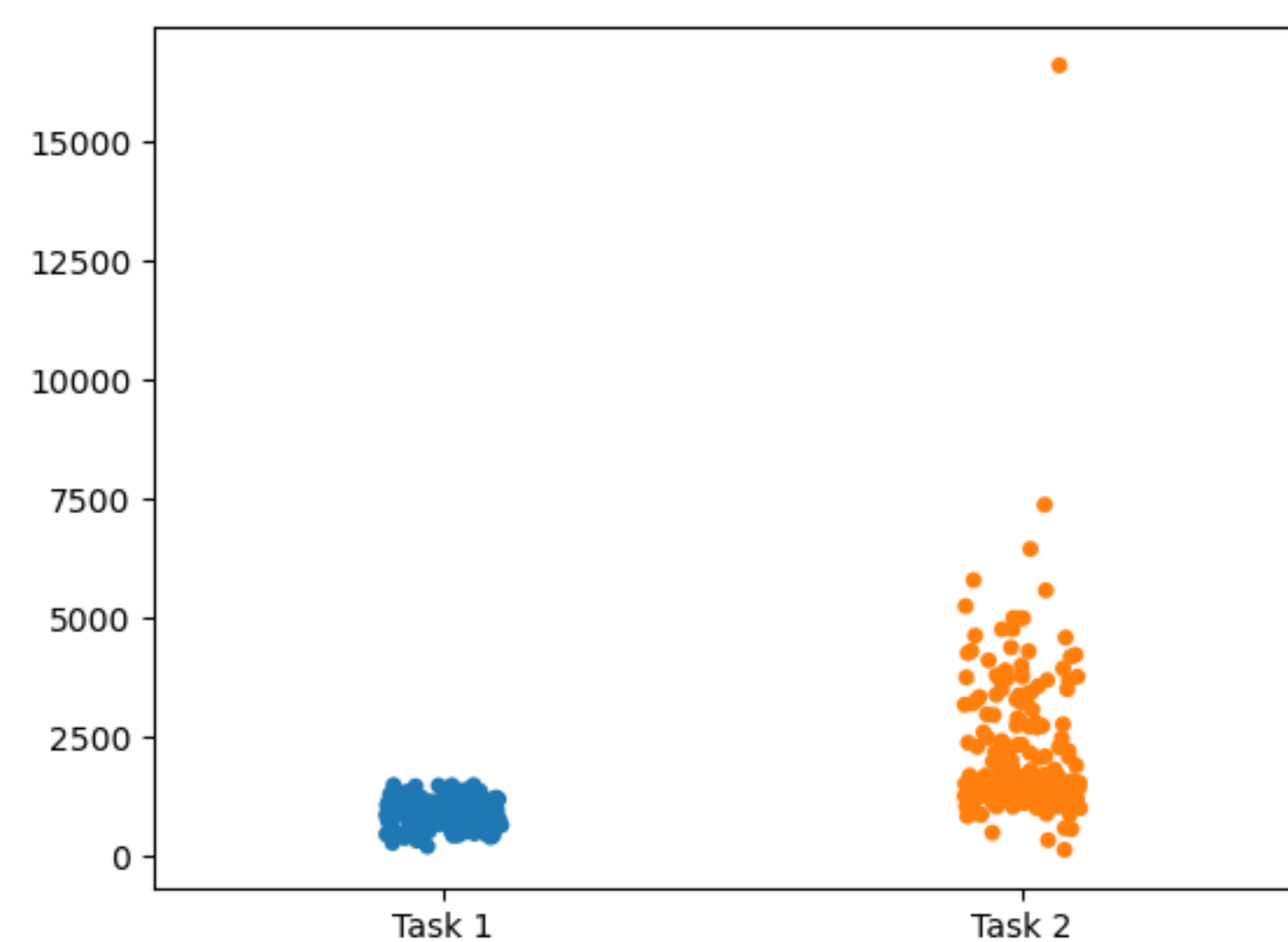


Figure 1: Text lengths per task

Approach

- Originality = opposite of similarity
- Average originality of a text = Average inverse similarity to all other texts
- But how to define similarity? 🤔🤔🤔

Similarity measures

Metric	Metric type	Data	Range
Jaro-Winkler similarity	Character-level	Pre-processed texts	[0,1]
Euclidian similarity	Character-level	Pre-processed texts	[0,1]
Compression similarity	Subword-level	Pre-processed texts	[0,1]
Jaccard similarity	Word-level	Pre-processed texts	[0,1]
Jaccard similarity	Bigram-level	Pre-processed text bigrams	[0,1]
Jaccard similarity	Trigram-level	Pre-processed text trigrams	[0,1]
Cosine similarity	Static embeddings	Term-Frequency Inverse-Document-Frequency representations of pre-processed texts	[0,1]
Cosine similarity	Static embeddings	Okapi BM25 representations of pre-processed texts	[0,1]
Cosine similarity	Static embeddings	Doc2Vec model (trained on pre-processed texts) embeddings of pre-processed texts	[-1,1]
Cosine similarity	Static embeddings	Doc2Vec model (trained on website dataset from Guzman and Li (2023)) embeddings of pre-processed texts	[-1,1]
Cosine similarity	Contextual embeddings	Alibaba-NLP/gte-large-en-v1.5 model embeddings of raw texts	[-1,1]
Cosine similarity	Contextual embeddings	mixedbread-ai/mxbai-embed-large-v1 model embeddings of raw texts	[-1,1]
Cosine similarity	Contextual embeddings	Alibaba-NLP/gte-Qwen2-7B-instruct model embeddings of raw texts	[-1,1]

Table 1: Similarity metrics used in this project

Website dataset source:
Guzman, J., & Li, A. (2023). Measuring founding strategy. *Management Science*, 69(1), 101-118. <https://doi.org/10.1287/mnsc.2022.4369>

Findings

a) Different similarity metrics = completely different results

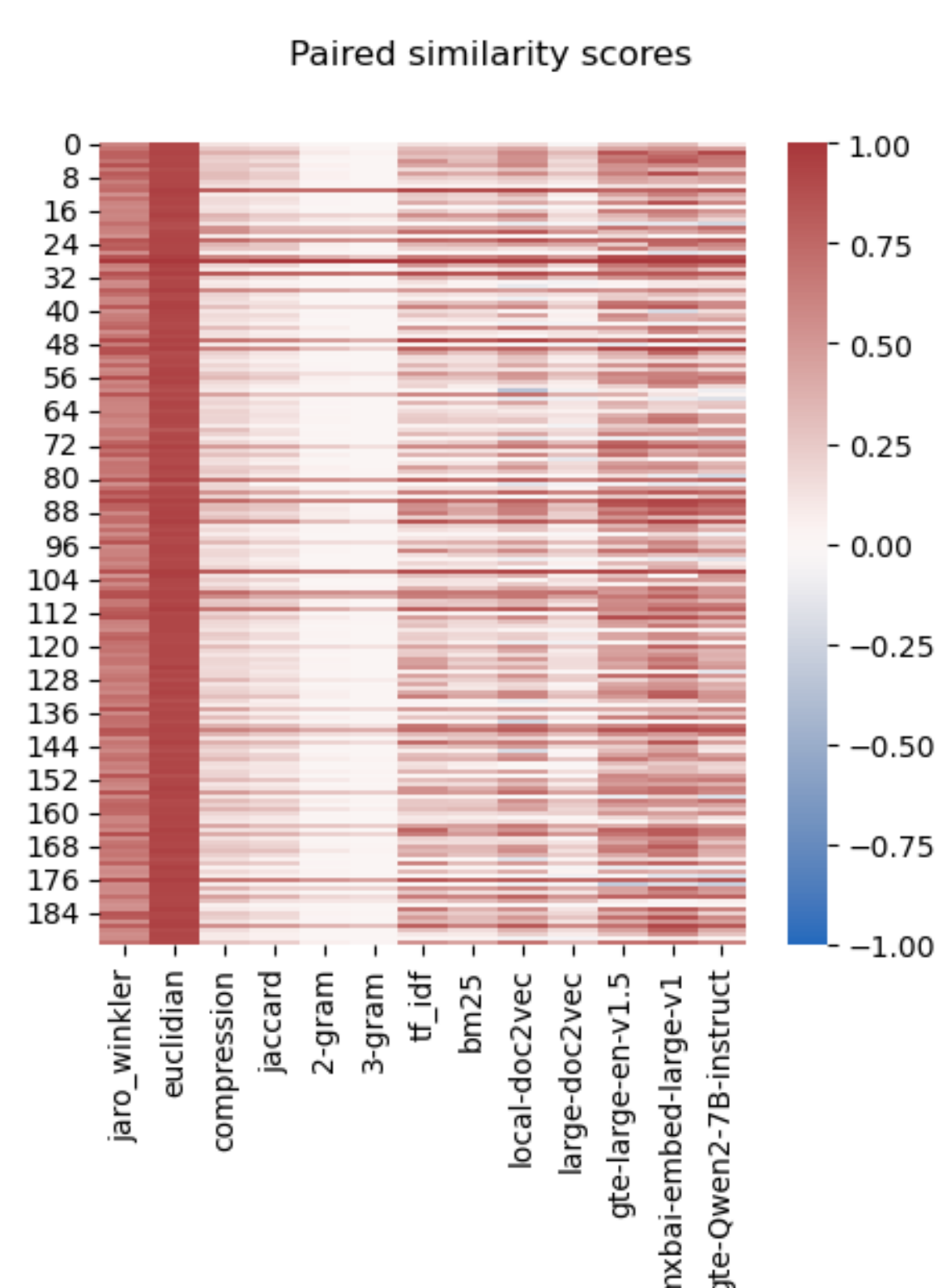


Figure 2: Visual heatmap representation of similarity scores for paired task 1 - task 2 comparisons

b) With composite metric (BM25 + Doc2Vec + gte-Qwen2-7B-instruct) -> statistically significant* but small differences in originality

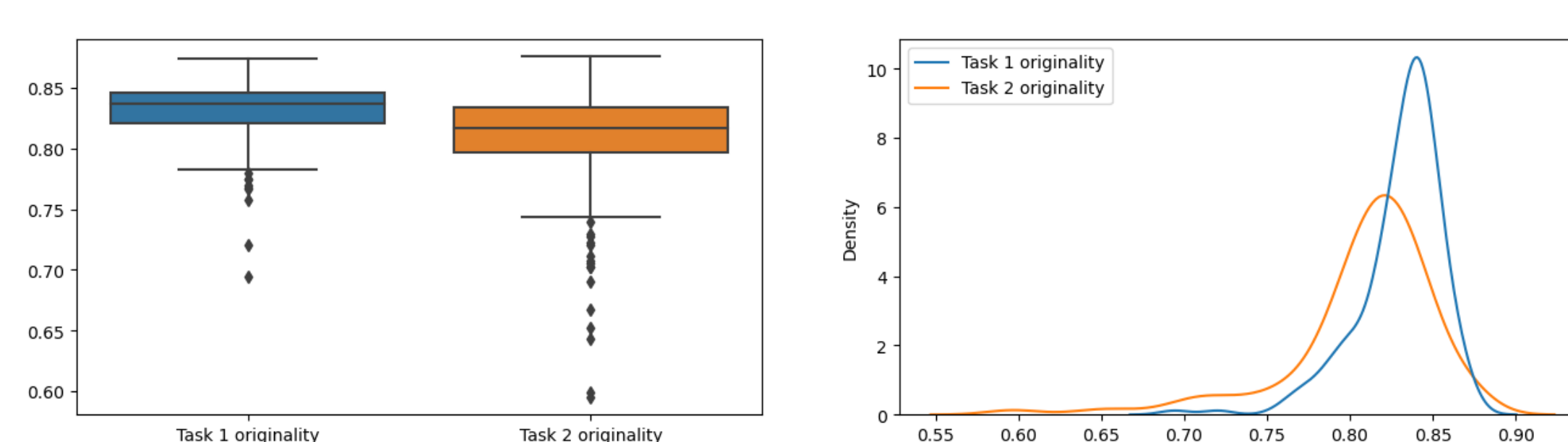


Figure 4: Final originality score distributions by task

*Wilcoxon signed-rank test with $p < 1e-9$

c) Embeddings models can help identify topic clusters and group insensible submissions together

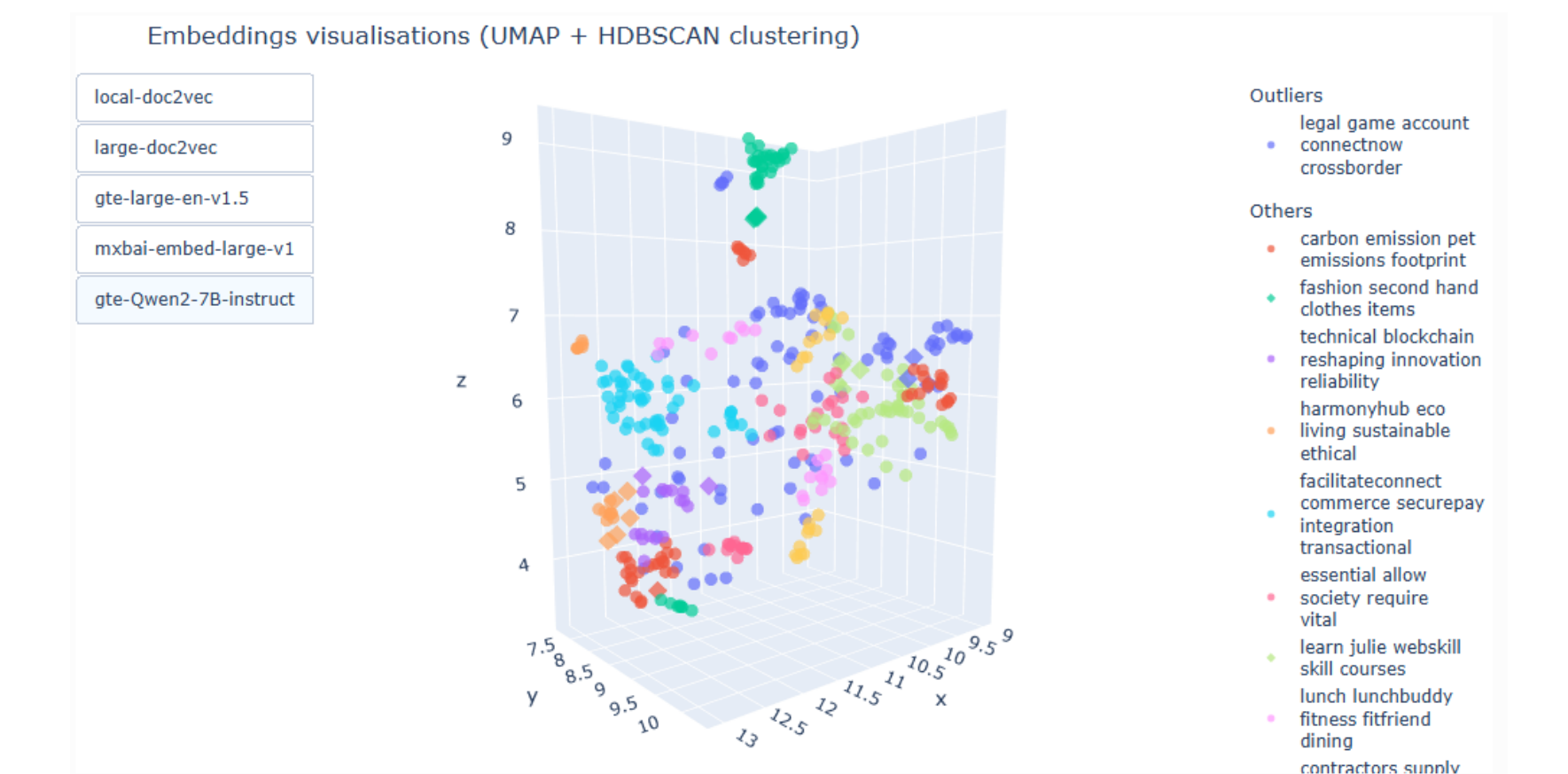


Figure 6: 3D visualisation of gte-Qwen2-7B-instruct embeddings of proposals coloured by topic cluster

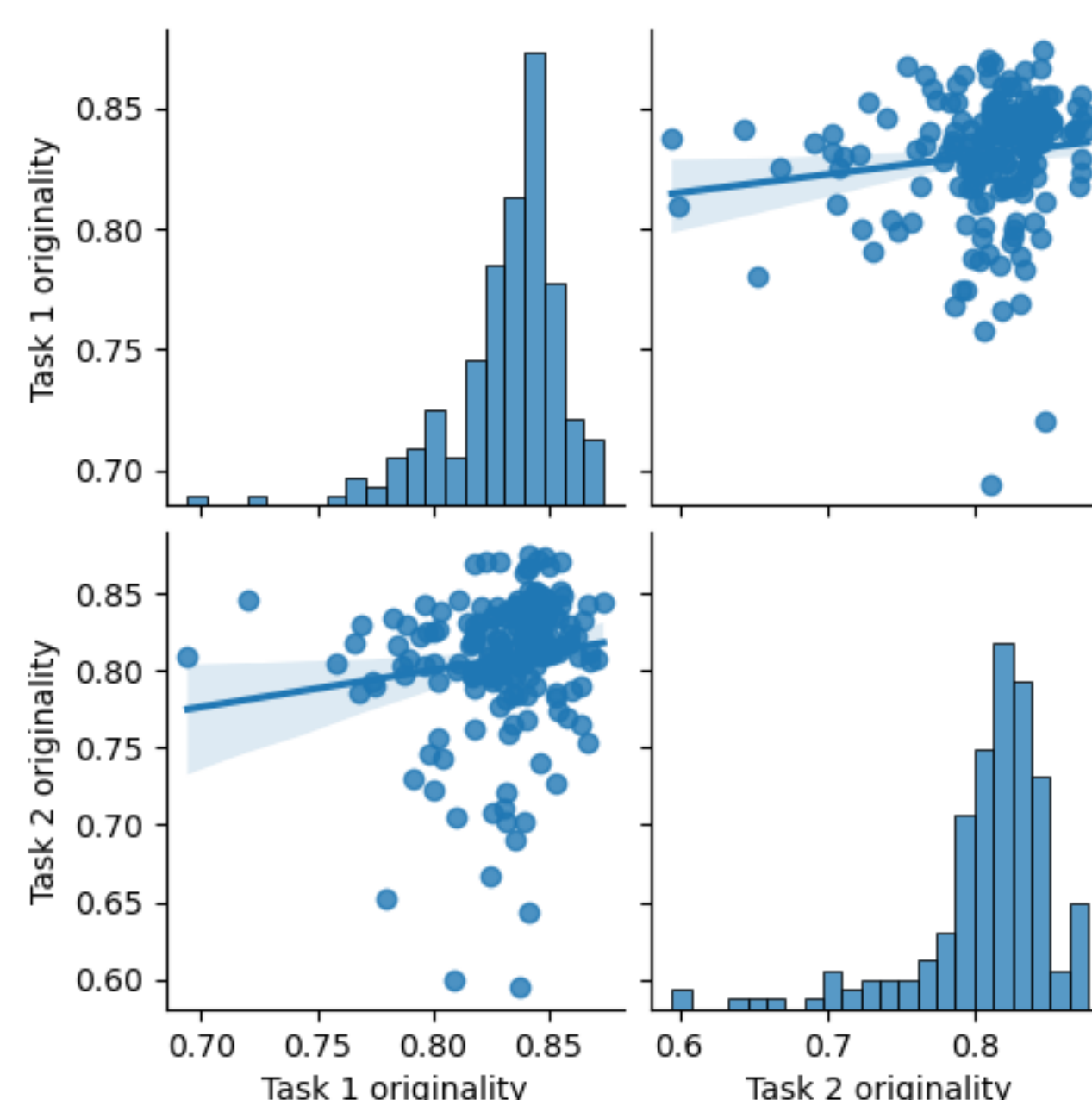


Figure 5: Linear regression of final originality scores by task

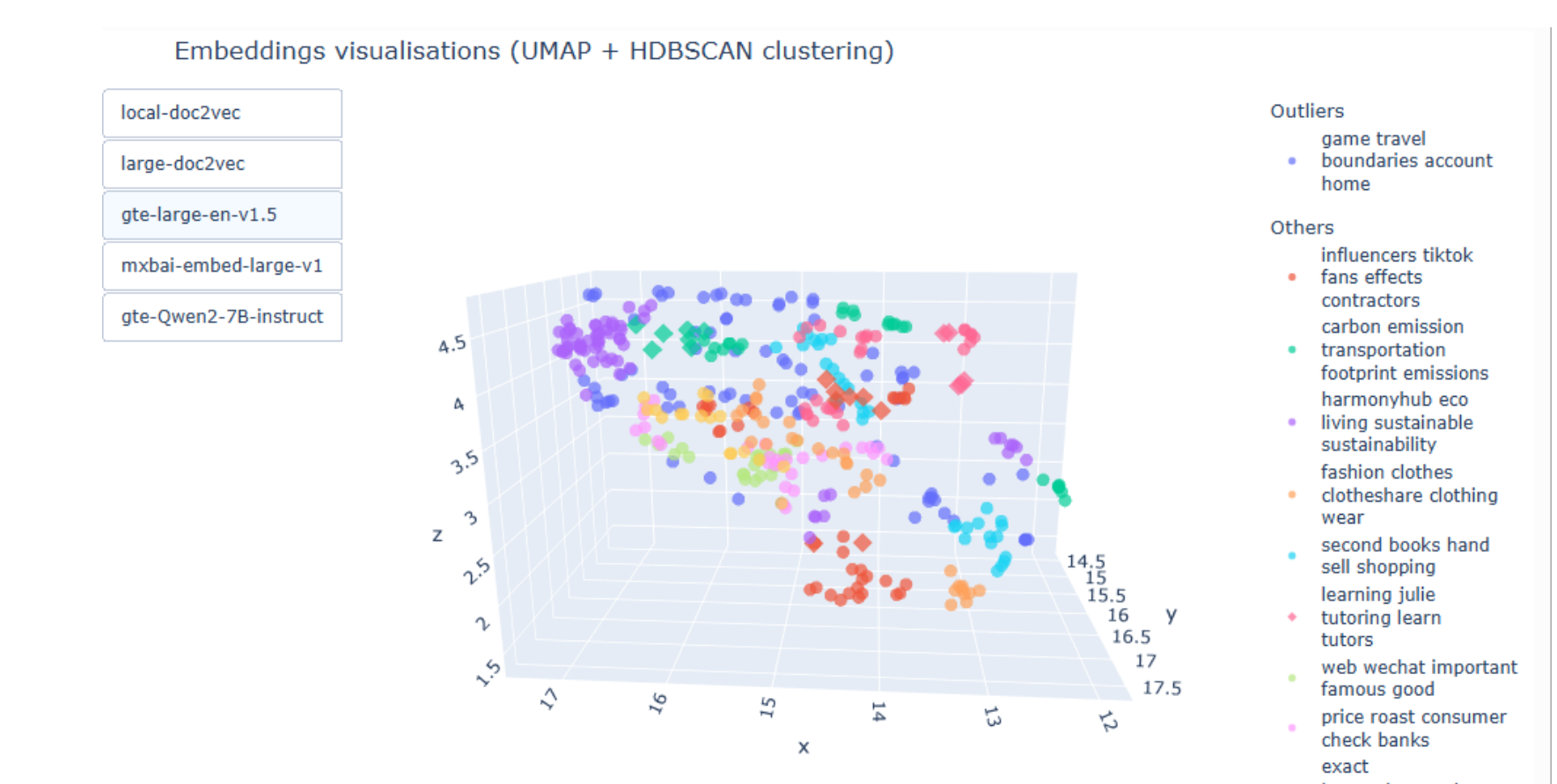


Figure 7: 3D visualisation of gte-large-en-v1.5 embeddings of proposals coloured by topic cluster

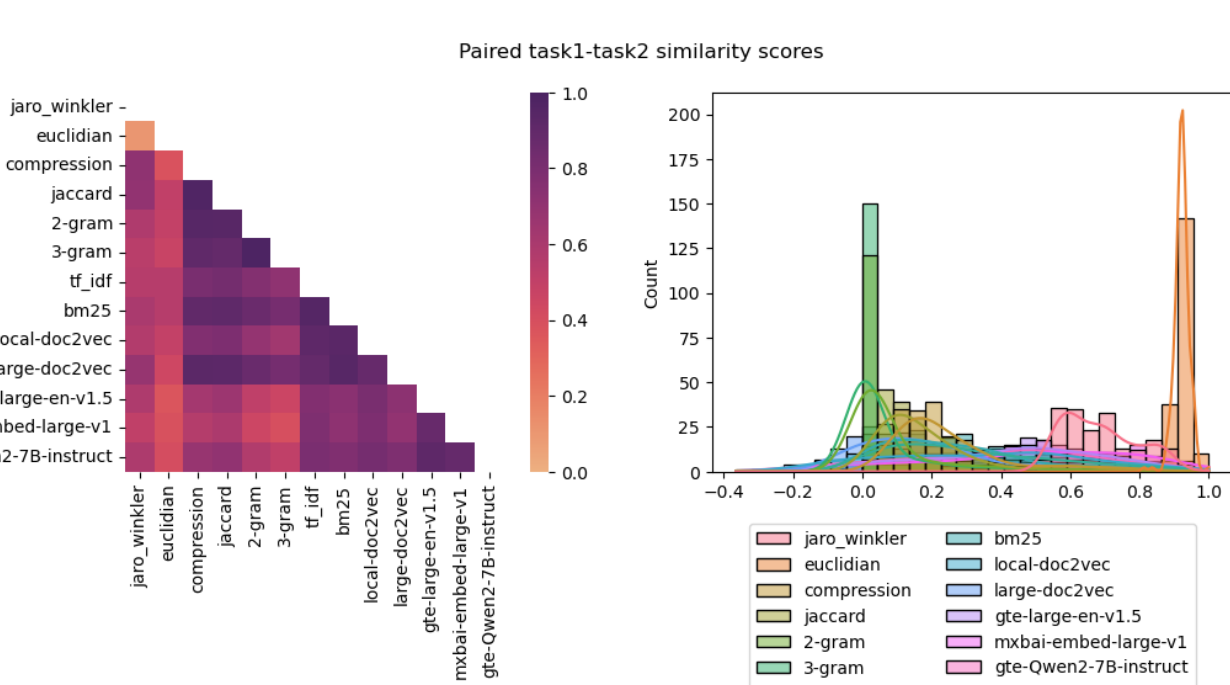


Figure 3: Correlations and distributions of paired task 1 - task 2 similarity scores using different measures

Next steps

Larger project will involve:

- Human marker annotation of originality, appropriateness, and types of interactions with ChatGPT.
- Human annotators will analyse a larger dataset that includes full ChatGPT chat histories.